THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL MCQ TEST SET 1

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The area of the triangle with vertices $(0,0)$, $(10,0)$ and $(0,14)$ is: 1

(a) 71 **(b)** 72 **(c)** 70 **(d)** 69

2. The area (in square units) of the triangle formed by the line $y = 2x + 10$ and the two axes is: 1

(a) 24 **(b)** 26 **(c)** 25 **(d)** 27

3. The value of $0.\overline{2} + 0.\overline{5}$ is: 1

(a) $\frac{2}{3}$ **(b)** $\frac{8}{9}$ **(c)** $\frac{7}{10}$ **(d)** $\frac{7}{9}$

4. If $x + y = 5$ and $xy = 6$, then $x^3 + y^3 =$ 1

(a) 37 **(b)** 35 **(c)** 36 **(d)** 34

5. Parallel chords of lengths 14 cm and 48 cm lie on opposite sides of the centre of a circle of radius 25 cm. The distance between them (in cm) is: 1

(a) 31 **(b)** 33 **(c)** 32 **(d)** 30

6. The area of a cyclic quadrilateral with sides 5 cm, 7 cm, 15 cm and 19 cm is (cm^2): 1

(a) 97 **(b)** 95 **(c)** 96 **(d)** 98

7. Two dice are rolled together. The probability that the product of the numbers is 12 is: 1

(a) $\frac{8}{9}$ **(b)** $\frac{2}{9}$ **(c)** $\frac{1}{9}$ **(d)** $\frac{1}{10}$

8. The 6th term of an AP is 20 and the 10th term is 32. The 16th term is: 1

(a) 53 **(b)** 47 **(c)** 56 **(d)** 50

9. The number of perfect squares between 100 and 1000 (both excluded) is: 1

(a) 20 **(b)** 23 **(c)** 21 **(d)** 22

10. A group of 34 people has average weight 111 kg. Another group has average weight 72 kg. Together the average weight is 78 kg. The number of people in the second group is: 1

(a) 186 **(b)** 188 **(c)** 189 **(d)** 187

11. Start with $n = 26$. Repeat: if n is even replace it by $n/2$, otherwise by $3n + 1$. The number of steps needed to reach 1 is: 1

(a) 9 **(b)** 11 **(c)** 10 **(d)** 12

12. The four angles of a quadrilateral are in AP with common difference 50°. The smallest angle is: 1

(a) 65° **(b)** 35° **(c)** 25° **(d)** 15°

13. The line $17x + 18y = 306$ cuts the axes at two points. The area of the triangle it forms with the axes (square units) is: 1

(a) 154 **(b)** 153 **(c)** 155 **(d)** 152

14. The diagonal of a cuboid measuring 12 cm $\times$ 24 cm $\times$ 24 cm is: 1

(a) 37 cm **(b)** 35 cm **(c)** 38 cm **(d)** 36 cm

15. The distance of the circumcentre of the right triangle with vertices $(0,0)$, $(6,0)$ and $(0,8)$ from the origin is: 1

(a) 7 **(b)** 4 **(c)** 6 **(d)** 5

16. The x-coordinate of the point where $y = 5x + 1$ and $y = 2x + 13$ meet is: 1

(a) 5 **(b)** 4 **(c)** 6 **(d)** 3

17. $0.\overline{13} + 0.\overline{31}$ equals: 1

(a) $\frac{1}{3}$ **(b)** $\frac{2}{5}$ **(c)** $\frac{5}{9}$ **(d)** $\frac{4}{9}$

18. If $x - y = 2$ and $xy = 15$, then $x^3 - y^3 =$ 1

(a) 100 **(b)** 99 **(c)** 98 **(d)** 97

19. Parallel chords of lengths 14 cm and 48 cm lie on the same side of the centre of a circle of radius 25 cm. The distance between them (in cm) is: 1

(a) 19 **(b)** 18 **(c)** 17 **(d)** 16

20. A circle of radius 7 cm (take $\pi = \frac{22}{7}$) is inscribed in a square. The area of the square not covered by the circle is (cm^2): 1

(a) 42 **(b)** 44 **(c)** 43 **(d)** 41

21. Two dice are rolled together. The probability that the difference of the numbers is 2 is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{7}{9}$ **(c)** $\frac{2}{9}$ **(d)** $\frac{1}{5}$

22. The sum of the first n terms of a sequence is $S_n = 2n^2 + 3n$. Its 9th term is: 1

(a) 41 **(b)** 37 **(c)** 39 **(d)** 35

23. For the expression $n^2 - n + 11$, the smallest natural number n for which the value is NOT prime is: 1

(a) 10 **(b)** 13 **(c)** 11 **(d)** 12

24. In what ratio must a 25% solution and a 41% solution be mixed to obtain a 34% solution? 1

(a) $9:7$ **(b)** $7:9$ **(c)** $8:9$ **(d)** $7:10$

25. Using Euclid's subtraction algorithm for gcd$(1344, 872)$, the number of subtraction steps needed is: 1

(a) 15 **(b)** 14 **(c)** 16 **(d)** 13

26. The sides of a rectangle are in the ratio $3:4$ and its diagonal is 25 cm. Its perimeter is: 1

(a) 75 cm **(b)** 105 cm **(c)** 70 cm **(d)** 95 cm

27. A taxi fare is ₹243 for 5 km and ₹425 for 12 km (fare is a linear function of distance). The fare for 38 km (in ₹) is: 1

(a) 1075 **(b)** 1101 **(c)** 1214 **(d)** 1127

28. Take $\pi = \frac{22}{7}$. A toy is a cone of height 6 cm placed on a hemisphere, both of radius 21 cm. Its total volume is: 1

(a) 24394 cm^3 **(b)** 22176 cm^3 **(c)** 19958 cm^3 **(d)** 26613 cm^3

29. A circle has diameter with endpoints $(1, -2)$ and $(19, 78)$. Its radius is: 1

(a) 40 **(b)** 41 **(c)** 43 **(d)** 42

30. If $p(x) = 2x + 3$, then $p(p(4)) =$ 1

(a) 27 **(b)** 25 **(c)** 24 **(d)** 26

31. The smallest natural number n for which $\frac{n}{120}$ has a terminating decimal expansion is: 1

(a) 2 **(b)** 3 **(c)** 4 **(d)** 5

32. If $x + \frac{1}{x} = 3$, then $x^3 + \frac{1}{x^3} =$ 1

(a) 18 **(b)** 19 **(c)** 17 **(d)** 20

33. Two circles of radii 13 cm and 15 cm intersect in a common chord of length 24 cm and their centres lie on opposite sides of it. The distance between the centres (in cm) is: 1

(a) 14 **(b)** 15 **(c)** 13 **(d)** 16

34. Two concentric circles have radii 14 cm and 21 cm (take $\pi = \frac{22}{7}$). The area of the ring between them is (cm^2): 1

(a) 772 **(b)** 770 **(c)** 769 **(d)** 771

35. Three coins are tossed. The probability of getting at least one head is: 1

(a) $\frac{3}{4}$ **(b)** $\frac{1}{8}$ **(c)** $\frac{7}{8}$ **(d)** $\frac{7}{9}$

36. The sum of all integers from 1 to 100 that are divisible by 6 is: 1

(a) 816 **(b)** 810 **(c)** 822 **(d)** 826

37. For the expression $n^2 + n + 191$, the smallest natural number n for which the value is NOT prime is: 1

(a) 2 **(b)** 4 **(c)** 3 **(d)** 5

38. A bus goes from A to B at 20 km/h and returns along the same road at 80 km/h. The average speed for the whole journey (km/h) is: 1

(a) 32 **(b)** 31 **(c)** 33 **(d)** 34

39. Using Euclid's subtraction algorithm for gcd$(912, 720)$, the number of subtraction steps needed is: 1

(a) 7 **(b)** 10 **(c)** 8 **(d)** 9

40. The diagonal of a square is $18\sqrt{2}$ cm. Its perimeter is: 1

(a) 80 cm **(b)** 72 cm **(c)** 108 cm **(d)** 90 cm

41. **Assertion (A):** The point $(-3, 4)$ lies in Quadrant II. 1

Reason (R): In Quadrant II the abscissa is negative and the ordinate is positive.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

42. **Assertion (A):** The zero of the polynomial $p(x) = 3x - 12$ is 4. 1

Reason (R): The zero of $ax + b$ $(a \neq 0)$ is $-\dfrac{b}{a}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

43. **Assertion (A):** The number 0.375 is a rational number. 1

Reason (R): Every terminating decimal can be written in the form $\dfrac{p}{q}$ with integers p, q and $q \neq 0$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

44. **Assertion (A):** $103^2 = 10609$. 1

Reason (R): $(a + b)^2 = a^2 + 2ab + b^2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

45. **Assertion (A):** In a circle of radius 5 cm, a chord at a distance of 3 cm from the centre has length 8 cm. 1

Reason (R): The perpendicular from the centre of a circle to a chord bisects the chord.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

46. **Assertion (A):** The area of a triangle with sides 13 cm, 14 cm and 15 cm is 84 cm^2. 1

Reason (R): By Heron's formula the area is $\sqrt{s(s-a)(s-b)(s-c)}$, where s is the semi-perimeter.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

47. **Assertion (A):** When a fair die is rolled, the probability of getting a prime number is $\frac{1}{2}$. 1

Reason (R): The prime numbers on a die are 2, 3 and 5, which are 3 of the 6 equally likely outcomes.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

48. **Assertion (A):** The sequence $2, 5, 8, 11, \ldots$ is an arithmetic progression. 1

Reason (R): The difference between consecutive terms is the constant 3.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

49. **Assertion (A):** The converse of 'If a number is divisible by 6, then it is even' is false. 1

Reason (R): The number 4 is even but is not divisible by 6.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

50. **Assertion (A):** Class A has 20 students with average 60 marks and Class B has 30 students with average 70 marks; the combined average is 66. 1

Reason (R): The combined average equals the total of all marks divided by the total number of students.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**